SEAT No.________ **No. of Printed Pages: 3**

[12]

SARDAR PATEL UNIVERSITY

BCA(SEMESTER-II) EXAMINATION ,2023

US02SBCA51/Mathematics

Date : 01/08/2023 Time: 2:00 PM To 4:00PM Total Marks:50

Q-1 Select Appropriate Answer. [10]

1 A Square matrix A is said to be skew symmetric if ____________

(a) $A \neq A^T$ (b) $A = -A^T$ (c) $A = A^T$ (d) None of these

2 Norm of the vector u = (3, 0, 4) is ______________

(a) 5 (b) 72 (c) 0 (d) 15

3 An Edge whose endpoints are the same vertex is called ______

(a) trivial graph (b) multigraph (c) loops (d) multiple

4 A vertex does not belongs to any edge is called ....

(a) trivial graph (b) isolated vertex (c) nullgraph (d)finitegraph

5 In a connected map with R =11, E = 27 then V = ________

(a) 18 (b) 19 (c) 36 (d) 20

6 A graph that can be drawn in a plane or on a sphere so that its edge do not cross is said to be_____-

(a) Complete (b) simple (c) planar (d) Non-planar

7 The value which occurs with the maximum frequency is called ____________

(a) Median (b) mode (c) mean (d) None

8 Mode of the observations 2, 5, 8, 4, 4, 5, 4, 6, 3, 4, 4 is ____________________

(a)3 (b) 7 (c) 4 (d) None

9 The median of 2, 4, 6, 1, 5, 3, 7 is______________

(a)3 (b) 4 (c) 5 (d) None

10 ____________ is the value dividing the observations in two equal parts.

(a)Mode (b) Median (c) Mean (d) None

Q-2 Short Questions(Attempt 8 out of 10) [16]

1 Find x and y if, x(1, 1)+ y(2, 1) = (1, 4)

2 If u = (1, 4, 3, 9), v = (-5, -2, 5, 6), then evaluate ||u +2v||.

(P.T.O.)

3 Find the value of k such that vectors (5, 4, k) and (3,-4, 1) are orthogonal.

4 If u = (1, 8, 3), v = (5, -2, 5), then evaluate (1) u+4v (2) u·v

5 Draw a diagram of the complete graphs K_6.

6 Draw a picture of each of the following graphs, and state whether or not it is simple.

G = (V1,E1), where V1 = {P,Q,R, S, T} and E1 = {PQ, PR, PS, PT, TR, PR}.

7 Draw a picture of each of the following graphs, and state whether or not it is simple.

G = (V1,E1), where V1 = {a, b, c, d, e} and E1 = {ab, bc, ac, ad, de}.

8 Explain the terms: Mean, Mode and median.

9 Find the Mean of 2, 4, 8, 12, 16 , 24, 42, 48 ,50

10 Find Mean of the following data:

X	0	1	2	3	4	5	6
F	15	21	25	43	51	40	39

Q-3 Find AB and BA for the following matrices [06]

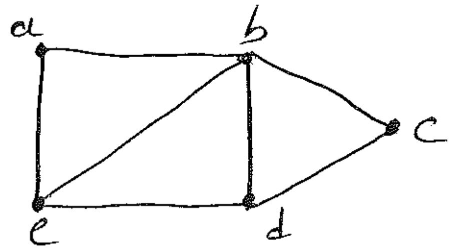
$$A=\begin{bmatrix} 1 & 2 & 3 \\ -2 & 3 & 4 \\ 1 & 0 & 1 \end{bmatrix} \quad B=\begin{bmatrix} 2 & 0 & 3 \\ 1 & 4 & 2 \\ 3 & 1 & 0 \end{bmatrix}$$

OR

Q-3 [06]

If $A=\begin{bmatrix} 1 & 3 \\ 4 & 2 \end{bmatrix}$ then evaluate f(A) for the polynomial $f(x) = x^3 + 3x^2 + 4x - 1$

Q-4 [06]



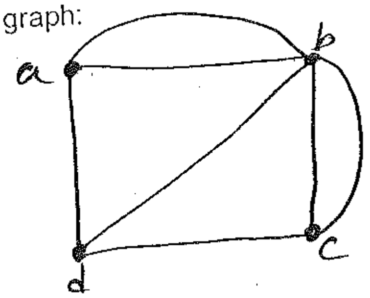
(a) Find cycle or closed path for given graph.

(b)Find the number m of edges in the graphs: 1) K_{120} 2) K_{200}

OR

[06]

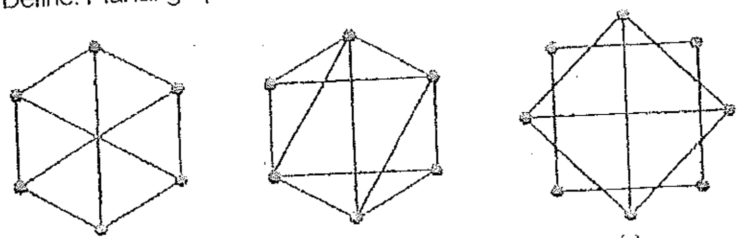
Q-4 For the following graph:



(a) Describe graph formally
(b) Find degree of all vertices and parity of each vertex.
(c) Is it true " The sum of each vertices is equal to twice the number of edges."

[06]

Q-5 Define: Planar graph. Checks which of the following are planar graphs. Justify.



(a) (b) (c)

OR

[06]

Q-5 The mean of 200 items was 50. Later on it was found that two items were wrongly read as 92 and 8 instead of 192 and 88. Find out the correct mean.

[06]

Q-6 Find the median & mode of the given data:

Marks	0-10	10-20	20-30	30-40	40-50	50-60
No. of Students	2	5	8	16	9	5

OR

[06]

Q-6 If the median of the following distribution is 38 find the missing frequency and number of observation are 300.

Class	10-20	20-30	30-40	40-50	50-60	60-70	70-80
Frequency	42	38	?	54	?	36	32

— x —